

05 Internetworking

05.03-05.04 DHCP and NAT

Introduction

- Dynamic Host Configuration Protocol is a communication protocol that lets network administrators manage centrally and automate the assignment of Internet Protocol addresses in an organization's network. (basically it frees a network administrator from having to manually configure each host connected to his network)

When and Who

- DHCP was created by the Dynamic Host Configuration Working Group of the Internet Engineering Task Force in the year 1987.
- It's definition was recorded in the RFC 1531 initially and was made obsolete by RFC's 1541 and 2131 subsequently.

About BOOTP

- DHCP is an extension of the Bootstrap protocol (BOOTP).
- BOOTP was designed for manual pre-configuration of the host information in a server database.

Purpose

- To deliver host-specific configuration parameters from a DHCP server to a host.
- Allocation of network addresses to hosts.

Overview

- DHCP is an application-layer protocol in the TCP/IP model.
- DHCP supports three mechanisms for IP address allocation.
 - Automatic allocation
 - Dynamic allocation
 - Manual allocation

Components of DHCP

- DHCP Server
- DHCP Client
- DHCP/BOOTP relay agent

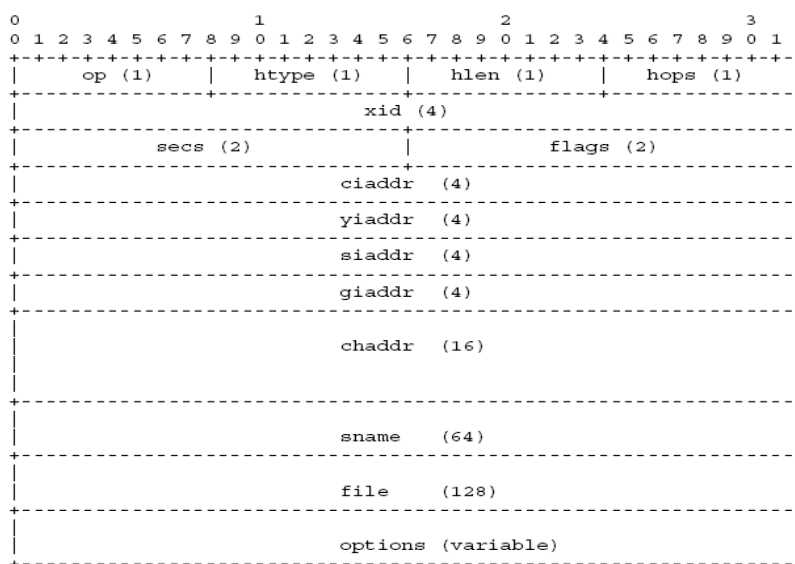
Design Goals

- Mechanism rather than a policy.
- Client should require no manual reconfiguration.
- It should not require a server on a subnet.
- It must interoperate with the BOOT relay agents and provide service to BOOTP clients.

DHCP must

- Guarantee unique network address.
- Retain DHCP client configuration across a client reboot.
- Allow automated assignment of configuration parameters to new clients.
- Support fixed allocation of configuration parameters to specific clients.

DHCP Message Format



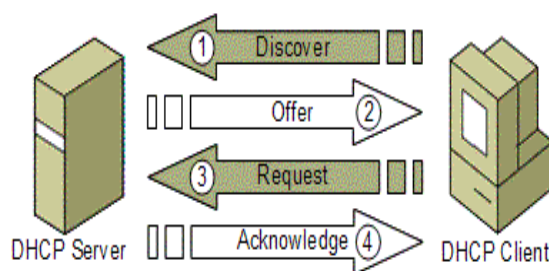
DHCP Messages

- DHCPDISCOVER - Client broadcast to locate available servers.
- DHCPOFFER - Server to client in response to DHCPDISCOVER with offer of configuration parameters.
- DHCPREQUEST - Client message to servers either (a) requesting offered parameters from one server and implicitly declining offers from all others, (b) confirming correctness of previously allocated address after, e.g., system reboot, or (c) extending the lease on a particular network address.
- DHCPACK - Server to client with configuration parameters, including committed network address.

DHCP Messages (..Contd)

- DHCPNAK - Server to client indicating client's notion of network address is incorrect (e.g., client has moved to new subnet) or client's lease as expired
- DHCPDECLINE - Client to server indicating network address is already in use.
- DHCPRELEASE - Client to server relinquishing network address and canceling remaining lease.
- DHCPINFORM - Client to server, asking only for local configuration parameters; client already has externally configured network address.

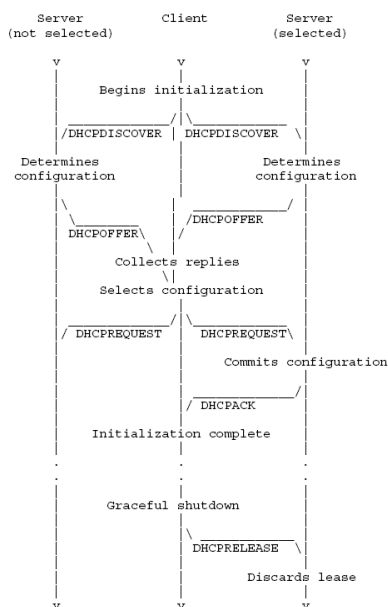
Dynamic allocation of a network address



Event Flow Diagram

- Two cases:
 - Client-server interaction while allocating a new network address
 - Client-server interaction while reusing previously allocated network address

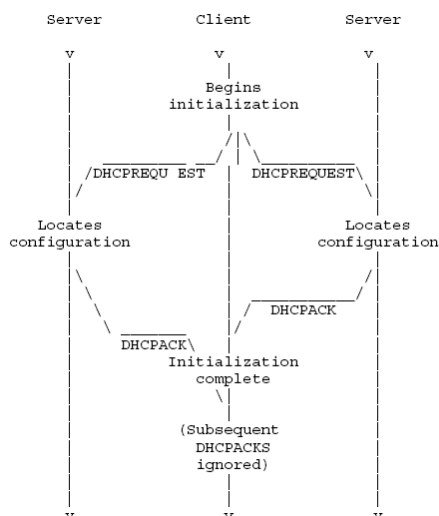
Allocating new network address



Step-by-step

- Client broadcasts DHCPDISCOVER
- Server respond with DHCPPOFFER
- Server check for address
- Client broadcasts DHCPREQUEST
- Server selected commits binding, sends DHCPACK
- Server sends DHCPNAK Client responds to DHCPACK
- Client sends DHCPDECLINE if address in use
- Client responds to DHCPNAK
- No DHCPACK/DHCPNAK
- Client sends DHCPRELEASE

Reusing previous network address



Step-by-step

- Client broadcasts DHCPREQUEST
- Server responds with DHCPACK
- Server SHOULD NOT check for network address
- Client is configured
- Client detects address in use sends DHCPDECLINE
- Client receives DHCPNAK
- No DHCPACK/DHCPNAK
- Client sends DHCPRELEASE

Steps-initialization with known network address

- Client begins in INIT-REBOOT state
- Sends DHCPREQUEST
- After DHCPACK, it moves to BOUND state

Reacquisition and expiration

- Times T1 and T2 to extend its lease
- At T1, client enters RENEWING state
- At T2, client enters REBINDING state
- $T1 < T2 < \text{lease expiration time}$
- In either of two states client sends DHCPREQUEST
- If lease expires before DHCPACK, client moves to INIT state

Enhancements

- Integration of DHCP with DNS
- Multicast Address allocation
- Unauthorized DHCP server detection
- Clustering for high availability

Performance Issues

- Appropriate lease durations
- Lengthening lease duration for large, fixed networks
- Shortening lease duration for variable networks with fewer IP addresses
- Reserve addresses with reservations
- Integrate DHCP with other services

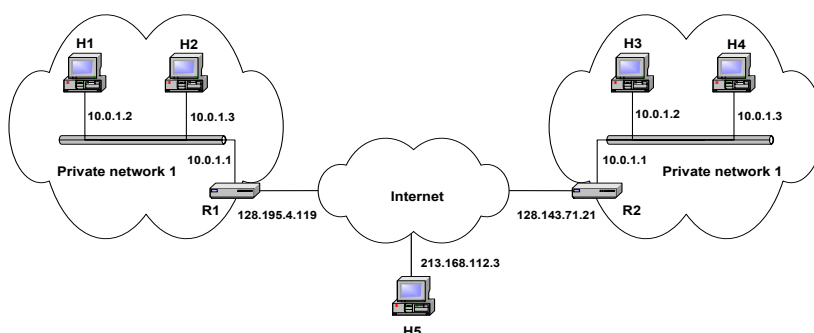
References

- RFCs 1531,1533,1541, 2131,2132, <http://ietf.org/rfc.html>
- DHCP FAQ
http://www.dhcp-handbook.com/dhcp_faq.html

Private Network

- *Private IP* network is an IP network that is not directly connected to the Internet
- IP addresses in a private network can be assigned arbitrarily.
 - Not registered and not guaranteed to be globally unique
- Generally, private networks use addresses from the following experimental address ranges (*non-routable addresses*):
 - 10.0.0.0 – 10.255.255.255
 - 172.16.0.0 – 172.31.255.255
 - 192.168.0.0 – 192.168.255.255

Private Addresses



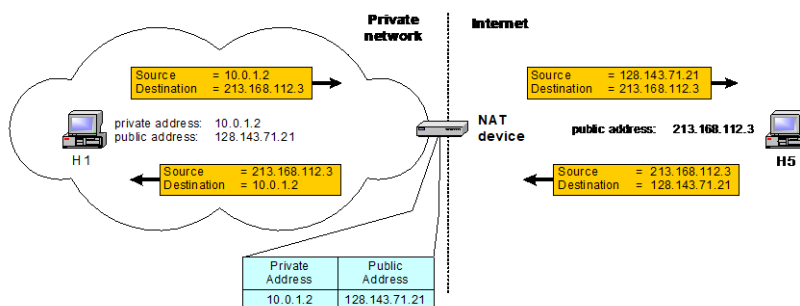
Network Address Translation (NAT)

- RFC 1631
- A short term solution to the problem of the depletion of IP addresses
 - Long term solution is IP v6
 - CIDR (Classless InterDomain Routing) is a possible short term solution
 - NAT is another
- NAT is a way to conserve IP addresses
 - Can be used to hide a number of hosts behind a single IP address
 - Uses private addresses:
 - 10.0.0.0-10.255.255.255,
 - 172.16.0.0-172.32.255.255 or
 - 192.168.0.0-192.168.255.255

Network Address Translation (NAT)

- NAT is a router function where IP addresses (and possibly port numbers) of IP datagrams are replaced at the boundary of a private network
- NAT is a method that enables hosts on private networks to communicate with hosts on the Internet
- NAT is run on routers that connect private networks to the public Internet, to replace the IP address-port pair of an IP packet with another IP address-port pair.

Basic Operation of NAT

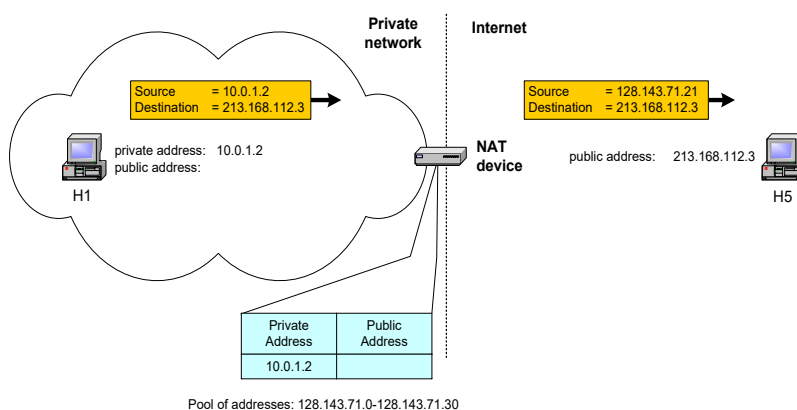


- NAT device has address translation table
- One to one address translation

Pooling of IP Addresses

- **Scenario:** Corporate network has many hosts but only a small number of public IP addresses
- **NAT solution:**
 - Corporate network is managed with a private address space
 - NAT device, located at the boundary between the corporate network and the public Internet, manages a pool of public IP addresses
 - When a host from the corporate network sends an IP datagram to a host in the public Internet, the NAT device picks a public IP address from the address pool, and binds this address to the private address of the host

Pooling of IP Addresses



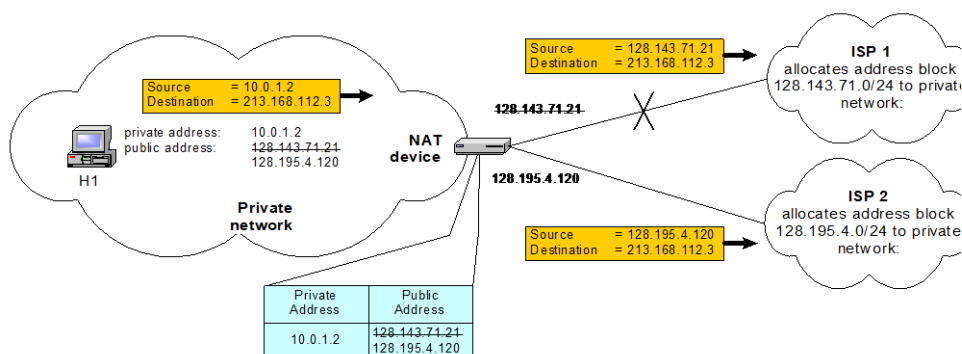
Supporting Migration between Network Service Providers

- **Scenario:** In CIDR, the IP addresses in a corporate network are obtained from the service provider. Changing the service provider requires changing all IP addresses in the network.
- **NAT solution:**
 - Assign private addresses to the hosts of the corporate network
 - NAT device has static address translation entries which bind the private address of a host to the public address.
 - Migration to a new network service provider merely requires an update of the NAT device. The migration is not noticeable to the hosts on the network.

Note:

- The difference to the use of NAT with IP address pooling is that the mapping of public and private IP addresses is static.

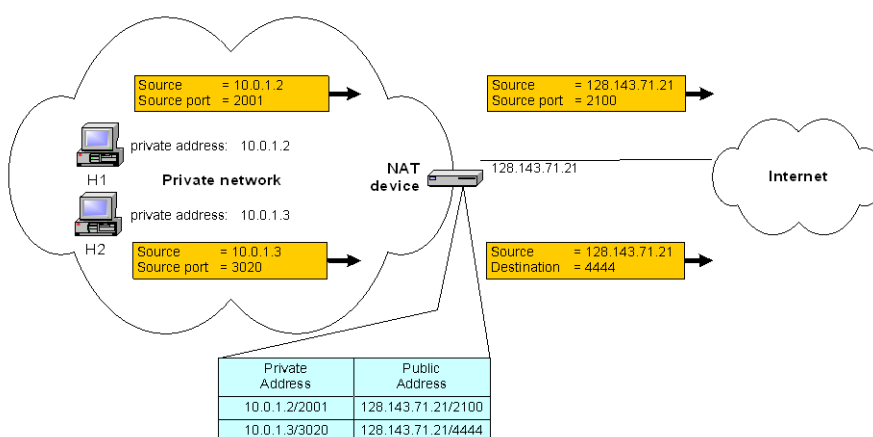
Supporting Migration between network service Providers



IP Masquerading

- **Also called:** Network address and port translation (NAPT), port address translation (PAT).
- **Scenario:** Single public IP address is mapped to multiple hosts in a private network.
- **NAT solution:**
 - Assign private addresses to the hosts of the corporate network
 - NAT device modifies the port numbers for outgoing traffic

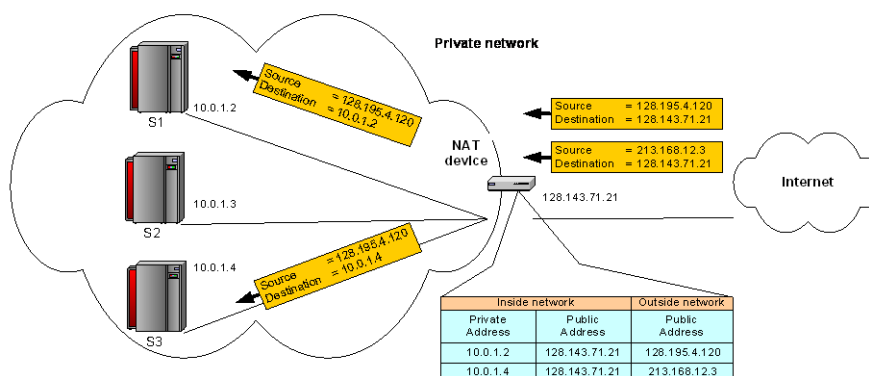
IP Masquerading



Load Balancing of Servers

- **Scenario:** Balance the load on a set of identical servers, which are accessible from a single IP address
- **NAT solution:**
 - Here, the servers are assigned private addresses
 - NAT device acts as a proxy for requests to the server from the public network
 - The NAT device changes the destination IP address of arriving packets to one of the private addresses for a server
 - A sensible strategy for balancing the load of the servers is to assign the addresses of the servers in a round-robin fashion.

Load Balancing of Servers



Concerns about NAT

- **Performance:**
 - Modifying the IP header by changing the IP address requires that NAT boxes recalculate the IP header checksum
 - Modifying port number requires that NAT boxes recalculate TCP checksum
- **Fragmentation**
 - Care must be taken that a datagram that is fragmented before it reaches the NAT device, is not assigned a different IP address or different port numbers for each of the fragments.

Concerns about NAT

- **End-to-end connectivity:**
 - NAT destroys universal end-to-end reachability of hosts on the Internet.
 - A host in the public Internet often cannot initiate communication to a host in a private network.
 - The problem is worse, when two hosts that are in a private network need to communicate with each other.



Concerns about NAT

- **IP address in application data:**
 - Applications that carry IP addresses in the payload of the application data generally do not work across a private-public network boundary.
 - Some NAT devices inspect the payload of widely used application layer protocols and, if an IP address is detected in the application-layer header or the application payload, translate the address according to the address translation table.

